Title: Project Outline - Document Q&A with source verification (RAG + citation tracking)

AI System Selection:

1. What AI system do you plan to design, develop, and experiment with?

- I plan to develop a Document Question-Answering system with Citation verification.
- The system will answer user questions using a collection of documents.
- It will implement Retrieval-Augmented Generation (RAG) to retrieve relevant documents before generating an answer.
- The AI system will provide citations of the document from which answer came.
- **Dataset used:** IBM Research watsonxDocsQA which contains corpus(1.14k rows) and document-question-answer pairs with citations(75 rows).
  (https://huggingface.co/datasets/ibm-research/watsonxDocsQA)

2. What AI concepts and techniques will you need to focus on?

- Retrieval-Augmented Generation (RAG) for searching the answer over the documents and generating the output.
- Vector embeddings to represent documents and queries in numeric format to understand the context(semantic search).
- Vector similarity search for retrieving relevant documents.
- Transformer-based foundation models for generating answers.
- Prompt engineering to guide the model to generate answers and also citations.
- Document indexing and search for efficient retrieval.
- Source attribution techniques to link answers to documents.

3. What computing resources, tools, platforms, datasets, existing implementations do you plan to use for your project?

- **Computing Resources:** VS Code, Python**.**
- **Tools and Platforms:** Hugging Face libraries for models and dataset loading, LangChain for implementing RAG pipelines, Chroma for storing document embeddings, Git and GitHub for version control.
- **Dataset:** The **watsonxDocsQA dataset** has real IBM product docs in text and markdown, like manuals and guides. It also has questions with answers and tells which documents the answers come from, so we can use it for RAG and citation tracking.

4. What risks might be posed by the development and use of the AI system? How can the risks be mitigated? Consider the following issues:

1. Safety and security:

Risks:

- The system might give wrong answers if it pulls irrelevant documents.
- Using private documents could leak sensitive info.

How to avoid:

- Stick to public datasets only.
- Always show the source documents with the answers.

2. Protection of data privacy and against algorithmic discrimination:

Risks:

- The dataset might have bias, which could affect the answers.
- Using confidential docs could create privacy problems.

How to avoid:

- Use only public research datasets.
- Check answers for fairness and accuracy.

3. Transparency:

Risks:

- Users might not know where the answer came from.

How to avoid:

- Show clear document-level citations.

4. Explanation:

Risks:

- Users might trust the answers without checking where they came from and does not give proper reasoning.

How to avoid:

- Always show the supporting document references.

Learning and Work Plan:

1. What's your learning and work plan for the duration of the project?

- Following the LLM course in Hugging Face spending 2-3 hours per week to better understand RAG, vector embeddings, semantic search.
- Learning to evaluate LLM performance in document QA tasks.

2. Map the project activities and work products on the project timeline.

WEEK	**Activities and products**
WEEK 7	• Project and Dataset Selection • Experiment with Dataset to understand the fields and the output format using print statements • Gather and Understand requirements • Submit Project plan
WEEK 8	• Set up development environment. • Load dataset and preprocess documents. • Generate document embeddings.
WEEK 9	• Implement vector database indexing using Chroma. • Build retrieval pipeline.
WEEK 10	• Integrate foundation model for answer generation. • Implement RAG pipeline. • Prepare Design Document
WEEK 11	• Implement citation tracking mechanism. • Ensure answers reference the source documents.
WEEK 12	Conduct system evaluation: • Retrieval accuracy • Answer correctness • Citation quality.
WEEK 13	• Testing
WEEK 14	• Documentation • Project presentation